

1. INGESTION TIERS

Tier 1: Cyberstation TUI

`k-cli ui` • Textual 60fps • 5-Persona Visual HUD • <160MB RAM

Tier 2: Cyber Station Web

`k-cli web-ui` • FastAPI SSE Real-time Token Streaming • Cyberpunk UI

Tier 3: Streamlined REPL

`k-cli chat` • Terminal Slash Commands • High-velocity scriptable pipe

Google Antigravity Runner

`k-cli exec <cmd>` • Non-blocking host shell execution engine

Background Healer Daemon

`k-cli daemon` • Continuous git branch watcher & proactive PR triage

2. INTENT SENSOR

Sub-0.1ms Heuristic Sensor

Zero-LLM overhead routing matrix with sub-millisecond classification

CHAT Intent

Direct low-latency streaming (<200ms) • Gemini 2.5 Flash

PLAN Intent

Milestone decomposition & architecture mapping • Nova Pro

BUILD / TRIAGE Intent

Closed-loop compiler verification pipeline • Strands Agent

IMMUNITY Intent

Adversarial chaos test generation & edge-case inoculation

Synapse AST Graph

In-memory symbol map of classes, functions, and cross-file dependencies

3. AWS STRANDS AGENT

StrandsDevAgent Core

Built with `from strands import Agent, tool` • Multi-turn reasoning loops

5-Persona State Machine

Researcher → Architect → Coder → Critic → Verifier consensus

Amazon Bedrock Models

- Amazon Nova Pro (`amazon.nova-pro-v1:0`)
- Google Gemini 2.5 Flash
- Local Sovereign Bankai SLMs (7B/14B)

Amazon Bedrock AgentCore

- 1-Click OpenAPI 3.0 Action Groups
- AWS SAM CloudFormation (`template.yaml`)
- AWS Lambda Runtime Bridge

4. DETERMINISTIC TOOLS

triage_and_heal_incident

7 RUNTIMES

verify_code_file

COMPILER

apply_surgical_patch

AST BLOCK

resolve_git_merge_conflict

3-WAY AST

execute_command

ANTIGRAVITY

generate_chaos_immunity_patch

INOCULATION

inspect_repo_structure

SYNAPSE

search_offline_docs

SQLITE FTS5

generate_architecture_diagram

MERMAID

Non-Blocking Subprocess Pool

Auto-injects virtualenv PATH (`sys.prefix/bin`) with real-time stream output

5. CLOSED-LOOP PROOF

1. AST Syntax Check

Validates Abstract Syntax Tree via `ast.parse` before touching disk

2. Native Compiler Proof

Invokes `py\_compile`, `g++`, or `cargo check` in isolated sandbox

3. Sandbox Pytest Run

Executes targeted regression tests to guarantee zero side effects

Self-Healing Retry Loop

If tests fail, compiler errors re-feed to agent for up to 3 automated repair iterations

Verified Delivery

- ✓ Staged Clean Surgical Diff
- ✓ AST Conventional Git Commit
- ✓ Zero Hallucinations Guaranteed